

FAISAL

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EDUCATION

University of Southern California
Master of Science in Computer Science
GPA: 4.0

Los Angeles, CA
August 2025-May 2027

Ramaiah Institute Of Technology (RIT)
Bachelor of Engineering in Information Science and Engineering
GPA: 9.41/10

Bengaluru, India
December 2020-June 2024

SKILLS

Languages and Frameworks: Python, Java, C/C++, JavaScript, React.js, React-Native, Flutter, Git, Pandas, Numpy, Flask, Matplotlib
Database and Tools: SQL, MySQL, MongoDB, Firebase, Postgres, GCP, AWS, Docker, Huggingface, HTML/CSS
Machine Learning/Artificial Intelligence: Tensorflow, Scikit-Learn, Llama Index, Langchain, LLMs, LSTMs, Neural Nets

EXPERIENCE

Melady Lab, University of Southern California
Graduate Research Assistant

Los Angeles, CA
May 2026-Present

- **TS-RAG:** Architected the time-series retrieval engine behind a system that sharpens forecasts using historically similar patterns (a vectorized, Parquet-backed similarity matcher spanning 16 domains and 33.4M observations), beating the prior engine head-to-head on 92% of queries.
- **CPT Pipeline:** Architected a config-driven ETL pipeline that built 500K time-series-text aligned records for continued pretraining(CPT) of a time-series foundation model across 14 domains.
- **TSFM Evaluation:** Architected a continuous-embedding baseline (frozen TimesFM-2.5 → Qwen3-1.7B, LoRA) testing whether an LLM reads time series as well from raw embeddings as from a discretized 1,000-token vocabulary.
- Built a reproducible harness benchmarking 8 LLM and time-series models across 3 suites (3,521 MCQ items, 355 forecasting instances), revealing that Llama-3.1-8B forecasts significantly worse when shown the series ($p < 0.05$).

DESE, Indian Institute of Science (IISc) & Boeing Research India
Research Staff

Bengaluru, India
August 2024-January 2025

- Designed a **risk-aware bi-directional A*** planner for Urban Air Mobility (eVTOL) routing that penalizes high-risk airspace using population, building, and weather data, cutting casualty-risk exposure ~30%.
- Added genetic-algorithm route deconfliction for multi-vehicle eVTOL traffic, resolving up to 95% of simulated route conflicts.
- Built an **eBPF/XDP** network-isolation layer that let an aircraft cabin display consolidate 3 separate backends (entertainment, service, lighting) onto one screen, verified by simulating DOS attack on one Docker-simulated backend while the other two remained 100% operational.

Indian Space Research Organisation
Research & Software Intern

Bengaluru, India
July 2023-May 2024

- Developed and benchmarked **unsupervised deep learning models** (LSTM, autoencoders) on 186k+ rows of multivariate satellite telemetry data (~40 features), boosting anomaly-detection F1-score by ~30% over statistical baseline models.

PROJECTS

Adaptive RAG for QA

May 2026

- Built an adaptive biomedical RAG router that decides between parametric **Llama 3.1 8B** and **FAISS retrieval** per question, lifting PubMedQA accuracy to 69% versus 53% parametric-only (78% ceiling for always-retrieving) after evaluating 3 router architectures (XGBoost, DistilBERT, LoRA).

LegendIDE (Winner at CalHacks 12.0)

October 2025

- Built a real-time, AI-powered web IDE that gives the AI **shared context across teammates'** code changes, intentions, and decisions; won Honorable Mention in the Letta track.
- Used **FastAPI/WebRTC** for session orchestration and Git ops, React for the UI, and a **CodeRabbit** → **Claude** → **Letta pipeline** for code-review context summarization.

PUBLICATIONS

Att-ConvGLUNet: A Novel Deep Learning Framework for Enhanced Brain Tumour Segmentation and Classification in MRI DICOM Imaging (Springer LNNS, 2026): 98% Tversky, 97% Dice on BraTS '23 segmentation.

Deciphering Gut Microbiome Dynamics in IBS using Deep Learning (Neurogastroenterology & Motility, 2025): ~93% accuracy classifying healthy vs. IBS-affected microbiomes.